SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Dharanik Kesh N.S (192494201) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dharanik Kesh N.S

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Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Q1: University Network Subnetting

1. Problem Understanding

A university has been assigned the IPv4 network: 192.168.20.0/24

The administrator wants to divide this network among three laboratories using different subnet sizes:

- Lab A → /26
- Lab B → /27
- Lab C → /28

We need to determine:

- Network address
- Valid host IP range
- Broadcast address
- Number of usable host addresses

2. Method Selection

For an IPv4 subnet with prefix /n:

Number of host bits = 32 − n

Total addresses = 2^(host bits)

Usable hosts = 2^(host bits) − 2

The subtraction of 2 accounts for the **network address and broadcast address**.

The block size can also be calculated as:

Block Size = 256 − subnet-mask value in the changing octet

3. Solution Process

Lab A – /26

Subnet mask: 255.255.255.192

Block size: 256 − 192 = 64

Therefore, the first subnet is:

- Network Address = 192.168.20.0

- First Host = 192.168.20.1
- Last Host = 192.168.20.62
- Broadcast = 192.168.20.63

Usable hosts:

2^(32–26) − 2

$= 2^6 - 2$

= 62 hosts

Lab B – /27

The next available address is 192.168.20.64.

Subnet mask:

255.255.255.224

Block size:

256 − 224 = 32

Therefore:

- Network Address = 192.168.20.64
- First Host = 192.168.20.65
- Last Host = 192.168.20.94
- Broadcast = 192.168.20.95

Usable hosts:

$2^5 - 2 = 30$ hosts

Lab C – /28

The next available address is 192.168.20.96.

Subnet mask: 255.255.255.240

Block size: 256 − 240 = 16

Therefore:

- Network Address = 192.168.20.96
- First Host = 192.168.20.97
- Last Host = 192.168.20.110
- Broadcast = 192.168.20.111

Usable hosts:

$2^4 - 2 = 14$ hosts

4. Final Results

	Prefix	Network Address	Valid Host Range	Broadcast	Usable Hosts
A	/26	192.168.20.0	.1 – .62	.63	62
B	/27	192.168.20.64	.65 – .94	.95	30
C	/28	192.168.20.96	.97 – .110	.111	14

5. Interpretation

The administrator can allocate different-sized subnets according to the requirements of each laboratory. Lab A receives the largest subnet with **62 usable hosts**, while Lab C receives the smallest with **14 usable hosts**. This avoids assigning the entire /24 network to one laboratory and improves IP address utilization.

Q2: Identifying Subnets of Two Systems

1. Problem Understanding

Given:

System 1: 192.168.1.130
System 2: 192.168.1.190

Subnet mask:

255.255.255.192 = /26

We need to determine:

- Subnet range
- Network address
- Broadcast address
- Valid host range
- Whether each IP is valid
- Number of usable hosts
- Whether both systems are in the same subnet

2. Method Selection

For /26:

Host bits:

$32 - 26 = 6$

Usable hosts:

$2^6 - 2 = 62$

Subnet mask:

255.255.255.192

Block size:

$256 - 192 = 64$

Thus, the subnet boundaries are:

- 0
- 64
- 128
- 192

3. Solution Process

Both IP addresses fall between 128 and 191.

Therefore, both belong to: 192.168.1.128/26

Subnet Details

- Network Address = 192.168.1.128
- First Host = 192.168.1.129
- Last Host = 192.168.1.190
- Broadcast = 192.168.1.191

System 1

IP: 192.168.1.130

Since: $129 \leq 130 \leq 190$

the IP is a valid host address.

System 2

IP: 192.168.1.190

Since: $129 \leq 190 \leq 190$

the IP is also a valid host address.

4. Final Results

Parameter	System 1	System 2
IP Address	192.168.1.130	192.168.1.190
Network	192.168.1.128/26	192.168.1.128/26
Host Range	.129 – .190	.129 – .190
Broadcast	.191	.191
Valid Host?	Yes	Yes

Usable hosts = 62

5. Interpretation

Both systems belong to the **same /26 subnet** and can communicate directly at the local network layer without requiring a router for communication between them, assuming normal LAN configuration.

Q3: Departmental Subnet Allocation

1. Problem Understanding

The organization has: **172.16.0.0/16**

Departments require:

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

We must determine the subnet address, valid host range, broadcast address, usable hosts, and remaining address space.

2. Method Selection

The departments are assigned different prefix lengths. Therefore, the allocation follows a **variable-sized subnet allocation approach**, where each department receives the specified subnet size.

Usable hosts:

2^(32−prefix) − 2

3. Solution Process

HR – /25

Host bits: **32 − 25 = 7**

Usable hosts: $2^7 - 2 = 126$

Network: 172.16.0.0/25

- Network = 172.16.0.0
- Hosts = 172.16.0.1 – 172.16.0.126
- Broadcast = 172.16.0.127

Sales – /26

Next address: 172.16.0.128

Host bits: $32 - 26 = 6$

Usable hosts: $2^6 - 2 = 62$

Therefore:

- Network = 172.16.0.128
- Hosts = 172.16.0.129 – 172.16.0.190
- Broadcast = 172.16.0.191

IT – /27

Next address: 172.16.0.192

Host bits: $32 - 27 = 5$

Usable hosts: $2^5 - 2 = 30$

Therefore:

- Network = 172.16.0.192
- Hosts = 172.16.0.193 – 172.16.0.222
- Broadcast = 172.16.0.223

Admin – /28

Next address: 172.16.0.224

Host bits: $32 - 28 = 4$

Usable hosts: $2^4 - 2 = 14$

Therefore:

- Network = 172.16.0.224
- Hosts = 172.16.0.225 – 172.16.0.238
- Broadcast = 172.16.0.239

4. Final Results

Department	Prefix	Network	Host Range	Broadcast	Usable Hosts
HR	/25	172.16.0.0	.1 – .126	.127	126
Sales	/26	172.16.0.128	.129 – .190	.191	62
IT	/27	172.16.0.192	.193 – .222	.223	30
Admin	/28	172.16.0.224	.225 – .238	.239	14

Remaining Address Space

Original /16 range: **172.16.0.0 – 172.16.255.255**

Allocated up to: **172.16.0.239**

Therefore, the remaining unused range is:

172.16.0.240 – 172.16.255.255

5. Interpretation

The allocation efficiently assigns larger subnets to departments requiring more hosts and smaller subnets to departments requiring fewer hosts. A large portion of the original /16 network remains available for future departments or network expansion.

Q4: Subnetting 172.16.10.0/24 Using /27

1. Problem Understanding

Given: **172.16.10.0/24**

New subnet mask: **/27**

We need to calculate:

- Number of subnets
- Usable hosts per subnet
- Subnet containing 172.16.10.78
- Network address
- Broadcast address
- Whether 172.16.10.95 belongs to the same subnet

2. Method Selection

Original prefix: /**24**

New prefix: /**27**

Number of borrowed bits: **27 − 24 = 3**

Number of subnets: $2^3 = 8$

Host bits: $32 - 27 = 5$

Usable hosts: $2^5 - 2 = 30$

3. Solution Process

Subnet mask: **255.255.255.224**

Block size: $256 - 224 = 32$

Subnets occur at: **0, 32, 64, 96, 128, 160, 192, 224**

The IP **172.16.10.78** lies between: **64 and 95**

Therefore, it belongs to: **172.16.10.64/27**

Details:

- Network = **172.16.10.64**
- First Host = **172.16.10.65**
- Last Host = **172.16.10.94**
- Broadcast = **172.16.10.95**

Checking 172.16.10.95

The address **172.16.10.95** is the broadcast address.

Therefore, it is within the subnet's address range but **cannot be assigned to a host**.

4. Final Results

- Number of subnets = **8**
- Usable hosts/subnet = **30**
- IP 172.16.10.78 → **172.16.10.64/27**
- Network = **172.16.10.64**
- Host range = **172.16.10.65 – 172.16.10.94**
- Broadcast = **172.16.10.95**
- 172.16.10.95 → **Same subnet, but broadcast address**

5. Interpretation

Subnetting the /24 network into /27 networks creates **8 smaller networks**, each capable of supporting **30 usable hosts**. The address .78 is a valid host, whereas .95 is reserved for broadcasting within that subnet.

Q5: IPv6 Address Compression and /64 Prefix

1. Problem Understanding

Given IPv6 address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Required:

- Compress the IPv6 address
- Expand it back
- Determine the /64 network prefix
- Calculate the number of possible addresses

2. Method Selection

IPv6 provides two important compression rules:

1. Leading zeros in each hexadecimal block can be removed.
2. One consecutive sequence of zero blocks can be replaced by ::.

An IPv6 address contains **128 bits**.

For /64: $128 - 64 = 64$ **host bits**

Therefore:

$$\text{Number of possible addresses} = 2^{64}$$

3. Solution Process

Original: **2001:0db8:0000:0000:0000:ff00:0042:8329**

Remove leading zeros: **2001:db8:0:0:0:ff00:42:8329**

Compress consecutive zero blocks:

Compressed form

2001:db8::ff00:42:8329

Expanded form

2001:0db8:0000:0000:0000:ff00:0042:8329

/64 Network Prefix

The first four groups represent the first 64 bits:

2001:0db8:0000:0000

Therefore:

Network Prefix = 2001:db8:0:0::/64

Number of Possible Addresses

Host bits: $128 - 64 = 64$

Therefore: $2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$

possible IPv6 addresses.

4. Final Results

Parameter	**Answer**
Original IPv6	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed	2001:db8::ff00:42:8329
Expanded	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix	/64
Network Prefix	2001:db8:0:0::/64
Host Bits	64
Possible Addresses	$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$

5. Interpretation

A /64 IPv6 network provides an extremely large address space of approximately **18.4 quintillion addresses**. The first 64 bits identify the network, while the remaining 64 bits are available for interface addressing. IPv6 does not use the IPv4 concept of a broadcast address.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately 2	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most 2 appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, 2.5 systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All 2 calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly 1.5 interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

	Problem understanding (20%)	method selection (20%)	Solution process (25%)	Accuracy of calculation (20%)	Interpretation of Results (15%)	
Q1	1.5	1.5	2	2	1	8
Q2	1	1.5	2	1.5	1	7
Q3	2	2	2.5	1.5	1	9
Q4	2	1	1	1	1.5	6.5
Q5	1.5	1.5	1.5	2	1.5	8

38.5